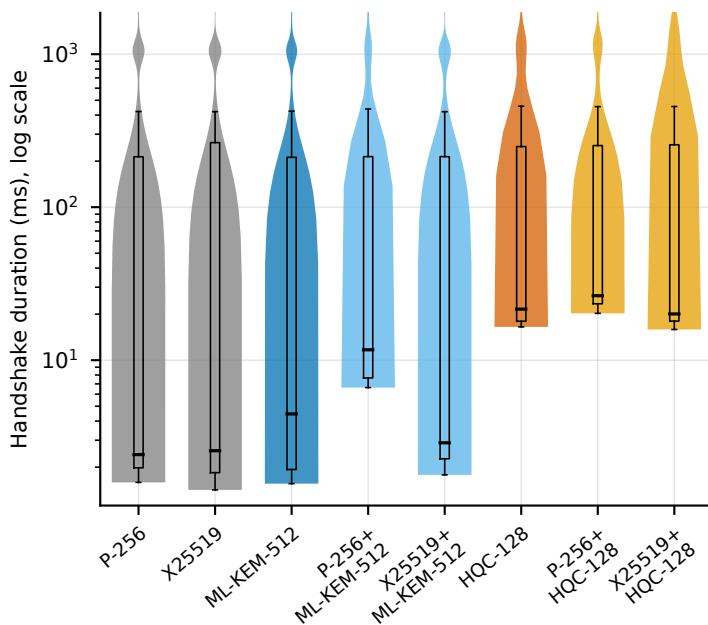
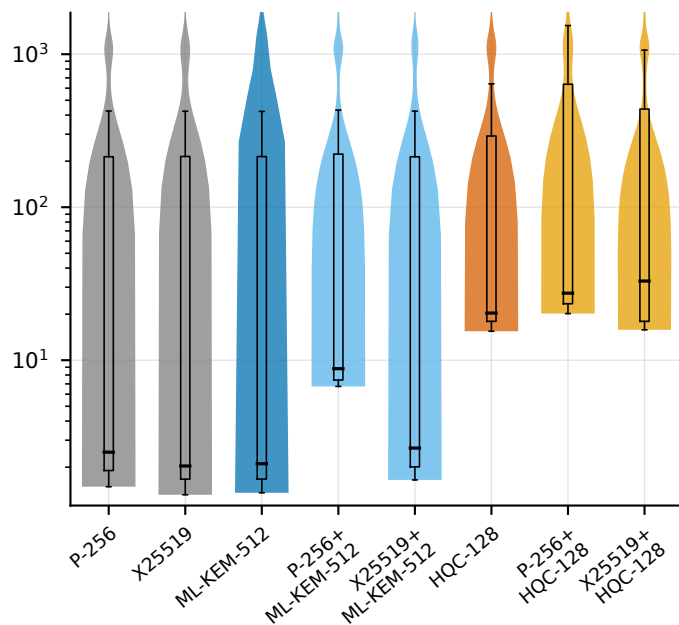


**(a)**

ed25519 (classical)

**(b)**

mlDSA44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

| KEM               | ed25519<br>Mean (ms) | ed25519<br>CV (%) | ed25519<br>Max (ms) | mlDSA44<br>Mean (ms) | mlDSA44<br>CV (%) | mlDSA44<br>Max (ms) |
|-------------------|----------------------|-------------------|---------------------|----------------------|-------------------|---------------------|
| P-256             | 292.7                | 169.0             | 2.67k               | 322.2                | 214.5             | 6.36k               |
| X25519            | 285.1                | 169.3             | 3.10k               | 328.9                | 235.3             | 6.62k               |
| ML-KEM-512        | 277.3                | 187.8             | 4.10k               | 384.3                | 352.8             | 26.35k              |
| P-256+ML-KEM-512  | 298.5                | 282.8             | 12.98k              | 330.7                | 186.0             | 3.34k               |
| X25519+ML-KEM-512 | 305.8                | 192.9             | 4.13k               | 306.4                | 215.3             | 6.37k               |
| HQC-128           | 367.8                | 249.6             | 14.36k              | 351.5                | 193.5             | 6.40k               |
| P-256+HQC-128     | 353.0                | 187.1             | 6.44k               | 472.9                | 179.8             | 6.43k               |
| X25519+HQC-128    | 423.7                | 329.2             | 27.23k              | 356.0                | 161.2             | 3.30k               |